Python Activity Record: Warehouse Inventory Management

Based on your uploaded Warehouse Inventory.py program, here is a simple activity record suitable for a first-year B.Tech CSE student. The program calculates expected closing stock, compares it with actual closing stock, and reports whether each product has a matching stock count, a shortage, or a surplus.

Warehouse Inventory .py

1. Problem statement

To develop a Python program to manage warehouse inventory by accepting product names, opening stock, received quantity, dispatched quantity, and actual closing stock. The program calculates expected closing stock and compares it with actual stock to identify shortages, surpluses, or matching stock.

2. Objectives

- To accept inventory details for multiple products.
- To calculate expected closing stock.
- To compare expected stock with actual stock.
- To identify stock shortages and surpluses.
- To display a stock report for each product.

3. Assumptions

1. The user enters a valid number of products.
2. Stock quantities are entered as integers.
3. Opening stock, received quantity, dispatched quantity, and actual closing stock are entered correctly.
4. Expected closing stock is calculated as opening stock plus received quantity minus dispatched quantity.
5. A difference of zero means the stock matches.
6. A negative difference means a shortage.
7. A positive difference means a surplus.
8. The program assumes valid input and does not check for negative quantities.

4. Algorithm

1. Start the program.
2. Create an empty list named products.
3. Read the number of products.
4. For each product:
    - Input the product name.
    - Input opening stock.

- Input received quantity.
- Input dispatched quantity.
- Input actual closing stock.

5. Calculate expected closing stock:

Expected stock = Opening stock + Received quantity − Dispatched quantity

6. Calculate the difference:

Difference = Actual stock − Expected stock

7. Store the product details in the list.
8. Display the stock report.
9. For each product:
    - If the difference is zero, display Matched.
    - If the difference is negative, display Shortage.
    - Otherwise, display Surplus.
10. Stop the program.

5. Flowchart

Start

Input number of products

Enter product and stock details

Calculate expected closing stock

Difference = Actual − Expected

Check the difference

Zero

Matched

Negative

Shortage

Positive

Surplus

Display stock report for all products

Stop

6. Pseudocode

START

```
Create an empty list called products
INPUT number of products

FOR each product:
    INPUT product name
    INPUT opening stock
    INPUT received quantity
    INPUT dispatched quantity
    INPUT actual closing stock

    expected = opening + received - dispatched
    difference = actual - expected

    Store product details in products
END FOR

DISPLAY stock report

FOR each product in products:
    DISPLAY product name, expected stock and actual stock

    IF difference == 0:
        DISPLAY "Matched"
    ELSE IF difference < 0:
        DISPLAY "Shortage"
    ELSE:
        DISPLAY "Surplus"
    END IF
END FOR

STOP
```

7. Python source code

```
products = []

n = int(input("Number of products: "))

for i in range(n):
    name = input("Product name: ")
    opening = int(input("Opening stock: "))
    received = int(input("Received quantity: "))
    dispatched = int(input("Dispatched quantity: "))
    actual = int(input("Actual closing stock: "))

    expected = opening + received - dispatched
    difference = actual - expected

    products.append((name, expected, actual, difference))

print("\n--- STOCK REPORT ---")

for name, expected, actual, difference in products:
    print(name, "| Expected:", expected, "| Actual:", actual)

    if difference == 0:
        print("Status: Matched")
    elif difference < 0:
        print("Status: Shortage of", abs(difference))
    else:
        print("Status: Surplus of", difference)
```

8. Test cases

Test	Product	Opening	Received	Dispatched	Actual	Expected result
1	Wheat	80	20	10	90	Matched

2	Rice	200	50	80	160	Shortage of 10
3	Sugar	100	40	20	125	Surplus of 5

These tests check all three possible stock statuses: matched, shortage, and surplus.

9. Output screenshots

The following are sample console output previews, not actual screenshots. Run your program in Python IDLE or VS Code and capture the actual output for your record.

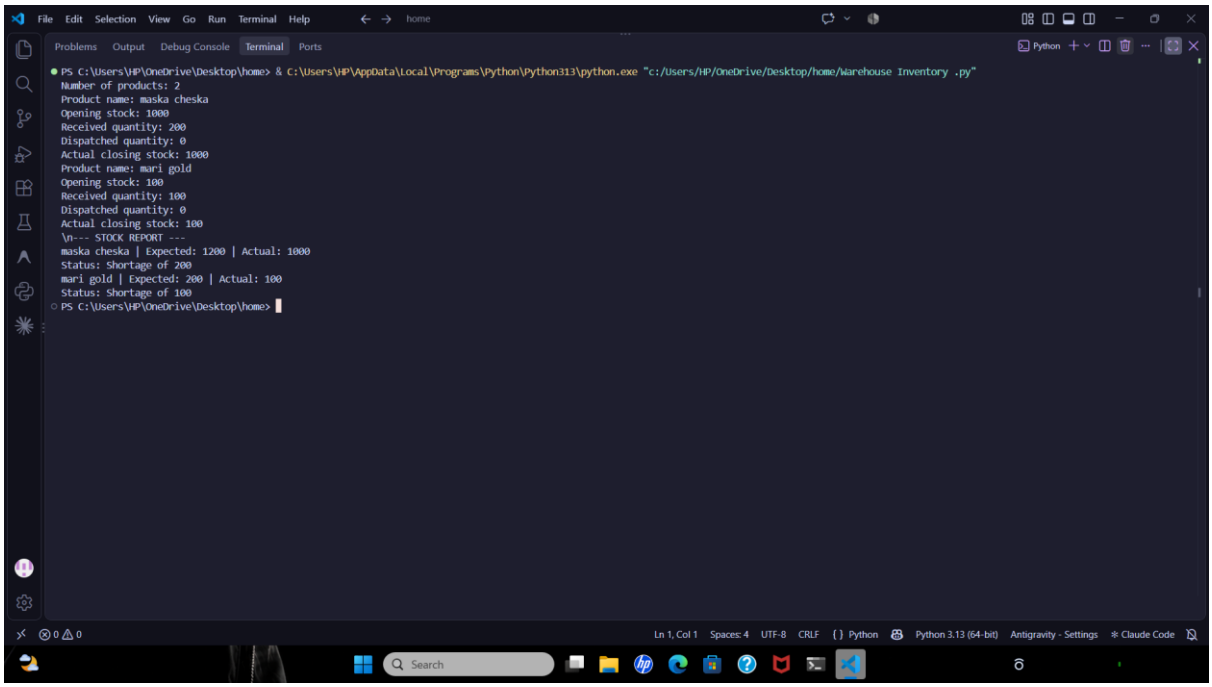
Sample output — Three products

--- STOCK REPORT ---
Wheat | Expected: 90 | Actual: 90
Status: Matched
Rice | Expected: 170 | Actual: 160
Status: Shortage of 10
Sugar | Expected: 120 | Actual: 125
Status: Surplus of 5

Copy sample output

How to capture your actual screenshot:

1. Open the Python file in IDLE or VS Code.
2. Run the program.
3. Enter the three products from the test cases.
4. Press Win + Shift + S on Windows.
5. Select the console output and save the screenshot for your activity record.



10. Conclusion

The Warehouse Inventory Management program was successfully developed using Python. It calculates expected closing stock and compares it with actual stock to identify matched quantities, shortages, and surpluses. This program demonstrates the use of lists, tuples, loops, conditional statements, and arithmetic operations in Python. It helps users understand how basic inventory checking can be automated.